PAPER_674: UQFF Compared to General LIGO Dataset: Strain, Phase, and Frequency Sweep

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Source: `grok_share_fc21e30c24b4.txt`
Domain: Gravitational Waves / LIGO

Abstract

We present the Unified Quantum Field Framework (UQFF) prediction for gravitational-wave strain $h_{\text{UQFF}}(f)$ across the LIGO sensitivity band (10–2000 Hz). The UQFF modifies the standard GR inspiral waveform through three multiplicative suppression factors: time-reversal zone f_{TRZ}, Schwarzschild-condensate S_{SCm}, and magnetic string modulation S_{U_m}. We demonstrate a systematic frequency-dependent suppression of order $\sim(1-f_{\text{TRZ}})\approx 0.9$ compared to GR, with phase shift $\Delta\phi=\kappa f_{\text{TRZ}} t_{\text{coal}}$.

Primary UQFF Equation

$$h_{\text{UQFF}}(f) = h_{\text{GR}}(f)\cdot(1-f_{\text{TRZ}})\cdot e^{-\rho_{\text{SCm}} r_s/k_B T_H}\cdot e^{-U_m 2\pi f/c^2}$$

Parameters

Parameter	Value	Description
m_1	$1.36\,M_\odot$	Primary mass
m_2	$1.17\,M_\odot$	Secondary mass
d	$40\,\text{Mpc}$	Luminosity distance
f	10–2000 Hz	LIGO band

UQFF Constants

Constant	Value	Description
ρ_{UA}	7.09×10^{-36} kg/m^3	Universal Aether density
ρ_{SCm}	7.09×10^{-37} kg/m^3	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor
κ	5×10^{-4} day^{-1}	UQFF calibration constant
$[\text{SSq}]$	0.57	Superstring quenching factor
μ_J	3.38×10^{23} J·m	Magnetic string coupling
γ	5×10^{-5} / 86400 s^{-1}	Aether oscillation decay

C++ Implementation

```cpp
#include "UQFFComparedToLIGOData.h"

UQFFComparedToLIGOData obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```python
from CondensedPhysics2 import UQFFComparedToLIGODataCalculator

calc = UQFFComparedToLIGODataCalculator()
result = calc.compute()  # returns all UQFF quantities
print(result)
```

References

LIGO Scientific Collaboration 2016 (GW150914); Abbott et al. 2019 (O3); UQFF: Session 173

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